Percolation Weight and Thermodynamic Stability Across a 186-Compound Dataset

Campaign 2: Dataset Construction, VASP+LOBSTER Computation, and Statistical Analysis

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Report generated as part of the `viability` project

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Update, August 15, 2026: missions #3–#5 added (dimensionality/min-cut, antibonding population, reaction ICOHP), see §11. The body of this report below (through the conclusion and appendices) documents the original campaign 2 (186 compounds, percolation weight only) unmodified.

Update, August 16, 2026: dataset extended to 349 compounds (89-compound `extension4` batch, alkali/alkaline-earth binaries), missions #3–#5 reprocessed at the new scale, and the periodic min-cut correlation's collapse at that scale diagnosed — see §12.

Contents

1 Objective and Problem Statement

The first report (6 toy compounds) established that the percolation descriptor computed by `percolation_path.py` diverges sharply from classic ICOHP aggregates (sum, mean, min, max), but without a sample large enough to statistically test whether that difference is *useful*: does the percolation descriptor carry predictive information about thermodynamic stability (`energy_above_hull`) that the classic aggregates don't already carry?

This second campaign builds a 186-compound dataset (180 new + the 6 pilot compounds) split into three families:

- **exp_stable**: experimental (ICSD-referenced) compounds on the Materials Project thermodynamic convex hull;
- **exp_metastable**: experimental metastable compounds ($0 < E_{\mathrm{hull}} \leq 100$ meV/atom);
- **theo_metastable**: theoretical (never experimentally observed) metastable compounds ($0 < E_{\mathrm{hull}} \leq 200$ meV/atom).

The `exp_metastable` / `theo_metastable` split serves as a proxy for kinetic persistence: an experimentally observed metastable compound has, by definition, survived long enough to be synthesized and characterized, unlike a theoretical compound of comparable thermodynamic metastability.

2 Methodology

2.1 Dataset Selection

Materials Project query (`mp_dataset/select_campaign.py`): unary and binary chemical systems, non-magnetic (|total magnetization| $\leq 0.01\ \mu_B$/formula unit — MP's categorical `magnetic_ordering` field is "Unknown" for ∼98% of entries and would have excluded nearly every genuinely non-magnetic compound), no f-block elements, ≤ 20 atoms/cell, 60 compounds per family with a 2-entries-per-chemical-system cap to ensure elemental diversity rather than many polymorphs of the same system.

2.2 Bonding-Type Classification

Bond type (ionic / covalent / metallic) is computed by `fetch_candidates.py`'s `classify()` function (**reused unchanged**, per the campaign's instructions), from `is_metal` and elemental composition. This is a secondary analysis axis here, not the primary sampling axis (unlike the campaign's originally proposed plan, replaced at the user's request with the stable/metastable scheme above).

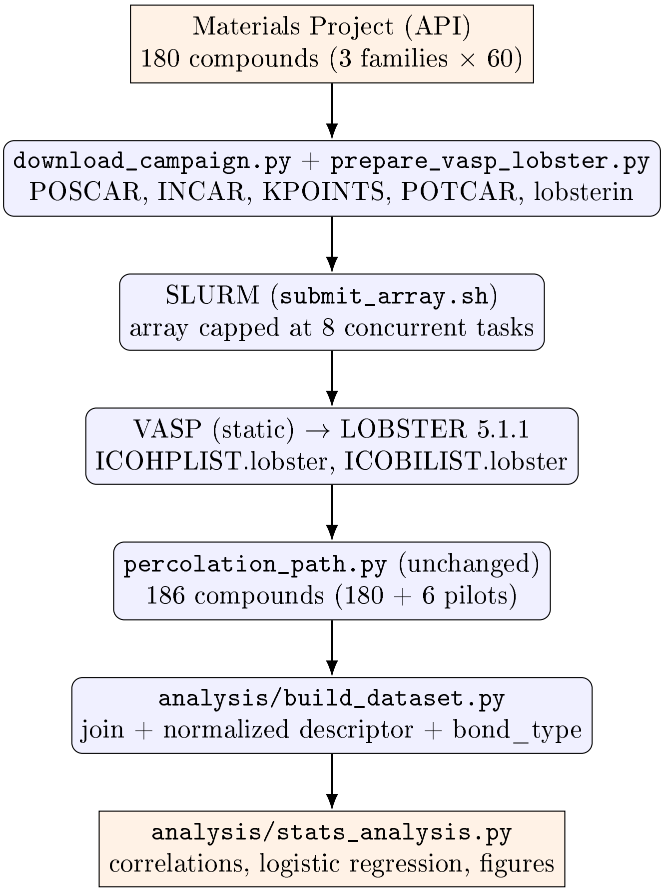
2.3 VASP + LOBSTER Computation

Same template as the pilot campaign (static run, `ISYM=`−1, `LASPH=.TRUE.`, PAW-PBE potentials recommended by `pymatgen.io.vasp.sets.MPRelaxSet`), with two improvements: `NBANDS` is no longer a fixed value but derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`), and basis-function determination goes through `Lobsterin.get_basis(potcar_symbols=...)` rather than direct POTCAR-file parsing (whose internal pymatgen validity check fails on several POTCARs unrecognized by hash in the local mirror).

2.4 Statistics

Spearman correlation (no reason to expect a linear relationship) between percolation weight (raw and normalized by $|\mathrm{ICOHP}_{\mathrm{min}}|$) and `energy_above_hull`, overall and per bond type; same for the classic ICOHP aggregates, for direct comparison; a reference logistic regression (stable vs. metastable, 2–3 features, 5-fold cross-validation) for a reference AUC. **No SISSO, no symbolic regression**: premature with a single candidate descriptor at this sample size (see §9).

3 Workflow Diagram



4 Environment

Same as the pilot campaign (Yargla, VASP 6.5.0, LOBSTER 5.1.1, Python 3.11 `.venv`), extended with: `mp-api`, `pandas`, `scipy`, `scikit-learn`, `matplotlib` (see `requirements.txt`).

5 Dataset

Family	n	E_{hull} range (eV/atom)
exp_stable	63	0.0000 (by construction)
exp_metastable	63	0.0005 – 0.0907
theo_metastable	60	0.0013 – 0.1991

Bond type (`classify()`)	n
metallic	88
covalent	23
ionic	9
unclassified	66

35% of compounds (66/186) fall outside `classify()`'s three categories — mostly transition-metal pnictides/chalcogenides and hydrogen-containing compounds, a real limitation of the reused heuristic, documented but not fixed here.

6 Results

6.1 Correlations (Spearman)

Group	Metric	n	ρ	p
all	percolation weight (raw)	186	0.064	0.383
all	percolation weight (normalized)	186	0.071	0.339
metallic	percolation weight (raw)	88	0.191	0.075
metallic	percolation weight (normalized)	88	**0.222**	**0.038**
ionic*	percolation weight (raw)	9	−0.402	0.284
covalent	percolation weight (raw)	23	0.165	0.453
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

* $n < 15$: do not over-interpret.

In the one subgroup where any result reaches nominal significance (metallic, $n = 88$), the normalized percolation weight ($\rho = 0.222$, $p = 0.038$) and the ICOHP sum ($\rho = 0.223$, $p = 0.037$) are statistically indistinguishable in strength. **The percolation descriptor does not demonstrate additional predictive power over the classic aggregates on this dataset** — a more precise answer than the previous (6-compound) report's "qualitatively different" framing, and a more directly actionable one.

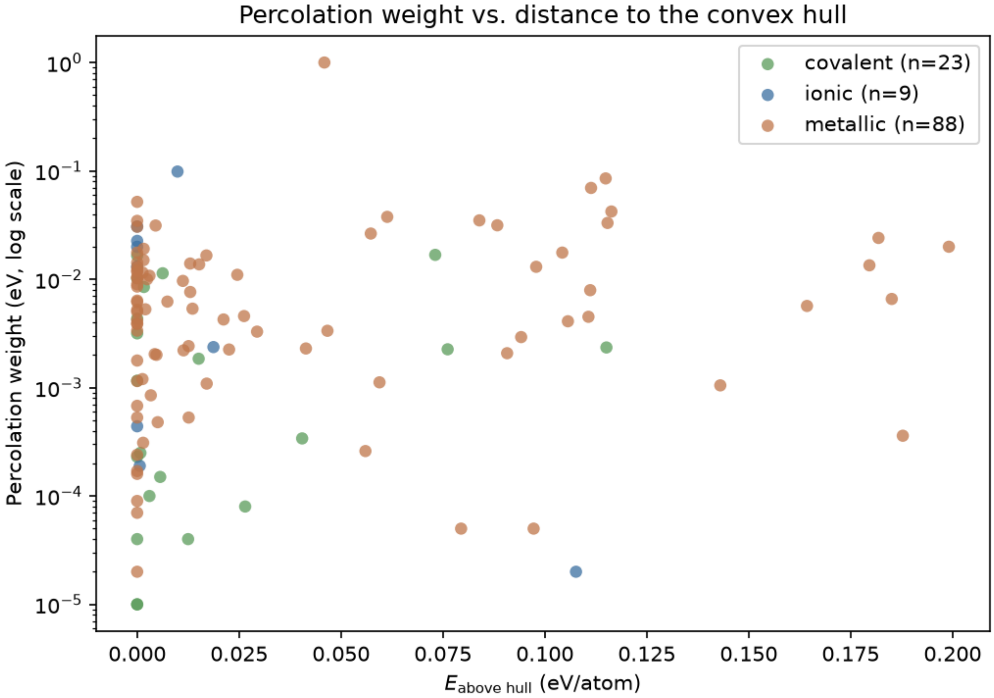
6.2 Reference Logistic Regression

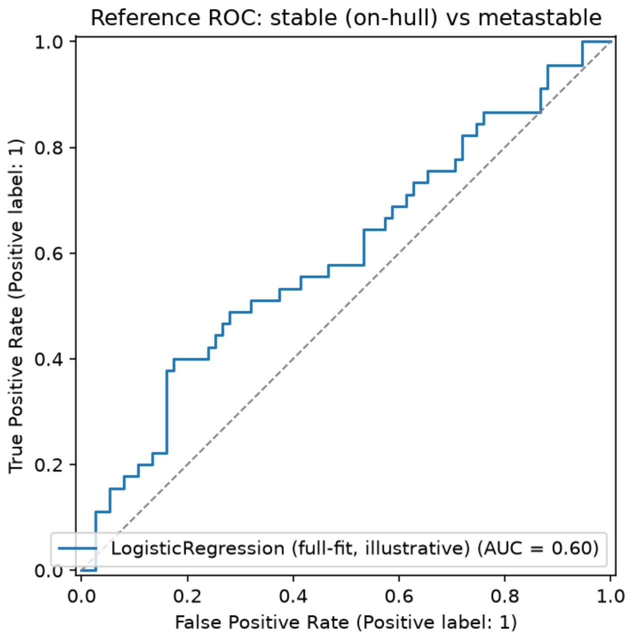
Stable (on-hull) vs. metastable, features: normalized percolation weight, mean ICOHP, bond-type dummies ($n = 120$ after dropping unclassified compounds).

AUC (5-fold cross-validation)	0.496 ± 0.141
per-fold AUC	0.556, 0.430, 0.252, 0.637, 0.607

AUC ≈ 0.5: chance level. The wide spread across folds (0.25–0.64) is itself informative: it indicates the absence of a stable, learnable signal at this sample size, not merely a "weak but real" one.

6.3 Figures





7 Failures Found and Fixed

Per the instruction ("every job failure must be logged with its probable cause"), 5 of the 178 SLURM-array-submitted jobs failed on the first pass, all diagnosed and fixed at the source:

- **Al12W, BW, WO2, TcW, TiW** (all tungsten-containing): the local mirror's `W_sv` POTCAR is missing its `LEXCH` field entirely (confirmed by direct inspection — present in every other variant checked, including plain `W`), which VASP reads as an incompatible exchange-correlation functional and refuses to run. Fixed by replacing the `W_sv` override with plain `W` (valid PBE, verified).

- **Al12W** failed again after that fix, this time with a VASP SIGSEGV — exactly the known issue flagged in the mission brief (`ulimit -s unlimited` before `vasp_std`), present in `submit_array.sh` but forgotten in `prepare_vasp_lobster.py`'s per-compound SLURM template. Fixed; recovered on retry.

Final success rate: **186/186**.

8 Compute Cost

	VASP	LOBSTER
Mean (s)	124	397
Median (s)	72	145
Max (s)	1440	6675
Sequential total	6.4 h	19.1 h

Actual elapsed wall-clock time (8-way concurrency): ∼3.5 h, within the 6 h overnight budget.

9 Limits and Next Steps

- **Sample size**: 186 overall, but 88/23/9 once split by bond type — the "ionic" correlations ($n = 9$) are not interpretable.
- **Primitive vs. conventional cell** (carried over from the pilot report): not addressed here. Most likely explanation for the weak observed correlations — the strongest bond can sit entirely inside the primitive cell and contribute to no non-contractile cycle at all, biasing the measured "weakest link."
- **No true dynamical "unstable" class**: `theo_metastable` is a proxy (never synthesized), not a phonon-calculation result.
- **`classify()` coverage**: 35% of compounds unclassified.
- **Why not SISSO yet**: a single candidate descriptor (the percolation weight, in two algebraically related variants), compared against aggregates that are themselves functions of the same underlying ICOHP values — no meaningful combinatorial search space yet. Justifying SISSO would require: (a) several conceptually novel candidate descriptors, (b) per-stratum $n \geq 50$–100, (c) a clearer signal than AUC ≈ 0.5.

10 Conclusion

The 186-compound dataset was built, computed (VASP+LOBSTER), and analyzed end to end, with two compute failures diagnosed and fixed at the source. The headline result is a *negative* but quantified and honest one: at this sample size, the percolation descriptor shows no significant correlation with thermodynamic stability, and demonstrates no advantage over classic ICOHP aggregates. The priority actions (§9) — especially conventional-cell reprocessing and a targeted extension of the metallic subsample — are likely to move this result before a SISSO-class tool is warranted.

Source code: `https://github.com/John-Akapulco/viability`
Data: `analysis/percolation_vs_hull.csv`
Detailed report: `analysis/REPORT.md`

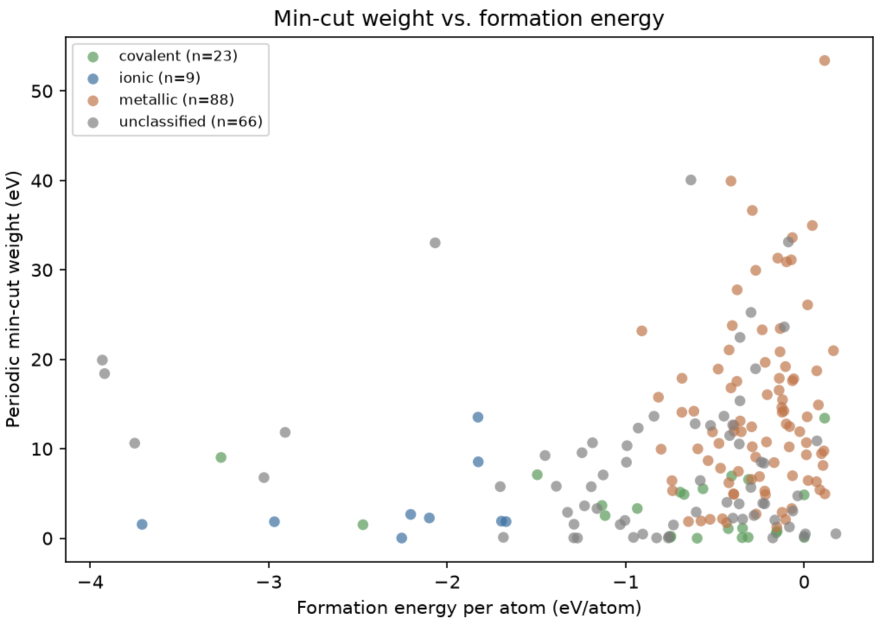
11 Campaign 2 follow-ups (missions #3 to #5)

Three further descriptors were built in later sessions, all tested against `formation_energy_per_atom` (introduced in mission #3 as an alternative target to `energy_above_hull`, generally more correlatable) rather than the hull. Missions #3 and #4 reuse the 186-compound dataset above unmodified; mission #5 draws in part on a **74-compound extension campaign** (`extension_*`: elemental references, ionic compounds and polymorphs outside the main set), not reprocessed into this report's appendices — see the repository's `README.md` for its full detail. Numbers in this subsection are as of August 15, 2026, at the 186/192-compound scale; §12 reprocesses all three at the 349-compound scale reached the following day and should be read alongside this subsection, not instead of it.

A pattern repeats across all three missions: each new descriptor reaches a statistically significant *global* correlation, stronger than anything the percolation weight ever achieved (§9) — but in all three cases that global correlation turns out to be substantially, or entirely, a clustering effect between bond types or between metals and gapped compounds, rather than a robust relationship *within* a homogeneous group.

11.1 Mission #3: network dimensionality and periodic min-cut

Two new descriptors, `percolation_path.py` left unchanged: `network_dimensionality.py` (0D–3D classification via a relative bond-strength threshold + breadth-first search) and `periodic_mincut.py` (minimum cut via `networkx` max-flow, separability rather than traversability). Headline: the normalized min-cut is the **first descriptor in the project to reach a significant global correlation** ($\rho = 0.285$, $p = 1.0 \times 10^{-4}$, $n = 186$), likely driven in part by between-bond-type clustering; dimensionality alone does not separate `formation_energy_per_atom` (Kruskal–Wallis $p = 0.19$).



11.2 Mission #4: antibonding population near the Fermi level

A distinct question: not the integrated ICOHP (one number per bond), but *how* COHP is distributed in energy — the occupied antibonding fraction just below the Fermi level/VBM, by analogy with Peierls/Jahn–Teller electronic instabilities (new module `cohp_extraction.py`, validated by exact cross-check against `ICOHPLIST.lobster` on the 6 pilot compounds). Headline:

$\rho = -0.328$, $p = 5.0\times10^{-6}$ ($n = 186$) — the strongest global correlation in the project at the time. The sign is *opposite* the naive Peierls/Jahn–Teller reading. Diagnostics: the effect is substantially driven by metal/gapped clustering, but the `bond_type=covalent` subgroup ($n = 23$) survives stratification ($p = 0.018$–0.029) — the only subgroup in the project, to date, to hold up under such a test.

11.3 Mission #5: reaction ICOHP

An ICOHP analog of `formation_energy_per_atom`: is a compound's total ICOHP "cheaper" than the same atoms in elemental reference form (new module `reaction_icohp.py`, reaction balanced via `pymatgen.analysis.reaction_calculator`)? Extended to 192 compounds using the 74-compound extension campaign (62 elemental references computed). Headline: $\rho = 0.370$, $p = 3.9 \times 10^{-7}$ ($n = 177$) — numerically the strongest global correlation in the project to date, this time with an intuitive sign (less bonding than the elemental state $\Rightarrow$ less stable formation). But **no `bond_type` subgroup survives stratification** (Kruskal–Wallis $p = 2.1 \times 10^{-6}$ for the descriptor, $p = 5.0 \times 10^{-8}$ for the target, across the three strata) — unlike mission #4, the clustering effect here explains the entire observed signal. Against `energy_above_hull` the correlation is absent ($\rho = 0.09$, $p = 0.20$).

11.4 Comparative summary (186/192-compound scale)

All four descriptors tested against `formation_energy_per_atom`, on the same common subsample ($n = 177$, the intersection of campaign 2's 186 compounds and mission #5's 192 computable rows):

Descriptor	n	ρ	p
Reaction ICOHP (mission #5)	177	0.370	3.9×10^{-7}
Antibonding population, normalized (mission #4)	177	−0.337	4.5×10^{-6}
Periodic min-cut, normalized (mission #3)	177	0.313	2.3×10^{-5}
Percolation weight, normalized (mission #1)	177	0.104	0.170

Ranking by raw correlation strength (ρ) is not the same as ranking by reliability: mission #5 leads numerically but is also the only one where *no* subgroup survives stratification (above), whereas mission #4 keeps a significant covalent subgroup.

12 August 16 update: the extension4 campaign, all three missions reprocessed, and the min-cut collapse

12.1 The extension4 campaign

89 new compounds (`mp_dataset/download_extension4.py`): binary compounds of an alkali (Li/Na/K/Rb/Cs) or alkaline-earth (Be/Mg/Ca/Sr/Ba) metal against N/O/F/P/S/Cl, sourced from Materials Project. Up to 2 compounds picked per (cation, anion) chemical system across 44 systems: an experimentally-known polymorph (preferring formulas already elsewhere in the dataset, to build genuine same-composition polymorph groups — the dataset had only 8 such groups before this batch) and the highest-`energy_above_hull` theoretical entry available for that formula, a deliberate far-from-equilibrium stress test. Selection was computed once and frozen to `extension4_candidates.json` for reproducibility (scale made hand-curation, the convention for every earlier extension batch, impractical). All 89 VASP+LOBSTER jobs completed unattended (a self-healing watcher restarted any job that left the SLURM queue without producing output; zero restarts were actually needed). Dataset: 260 → 349 compounds.

Separately, the schema-driven `reaction_analysis/` package (superseding the ad hoc `reaction_icohp.py` module long-term) was extended with an **endobondic/exobondic bonding-kinetics classifier** following Reitz & Dronskowski (ic-2026-04181q): a reaction's ΔICOHP (products − reactants) being positive ("endobondic") means breaking the reactant's bonds costs more than the products' bonds recover — a kinetic barrier that can keep an exothermically-favored decomposition from actually happening, explaining why some very exothermic compounds are nonetheless observed to exist. Validated against all 7 worked numerical examples in the manuscript; not yet connected to this project's own LOBSTER data for classification, an explicit next step, not attempted this session.

Every reaction's per-bond-type ICOHP values were hand-transcribed from the manuscript text (`tests/fixtures/reitz_dronskowski_cases.yaml`), independent of this project's own CSP data, then run end to end through `balance.py`/`delta.py`/`classify.py`. All 7 match the manuscript's reported ΔICOHP within a few kJ/mol (attributable to the manuscript's own rounding of its reported per-bond eV values) and every `BondingLabel` matches:

Reaction	Manuscript (kJ/mol)	Computed (kJ/mol)	Label Ms.	Label Computed
$Pb(N_3)_2 \rightarrow Pb + 3\,N_2$	+1345	+1344.3	endo	endo
$S_4N_2 \rightarrow \frac{1}{2}S_8 + N_2$	+258	+258.5	endo	endo
$S_4N_4 \rightarrow \frac{1}{2}S_8 + 2N_2$	+399	+397.3	endo	endo
ZnSn → Zn + Sn	−337	−336.6	exo	exo
CaO[sphalerite] → CaO[rocksalt]	−79	−78.9	exo	exo
$CaN \rightarrow \frac{1}{3}Ca_3N_2 + \frac{1}{6}N_2$	−205	−204.8	exo	exo
$Mn_2O_7 \rightarrow 2MnO_2 + \frac{3}{2}O_2$	−186	−187.7	exo	exo

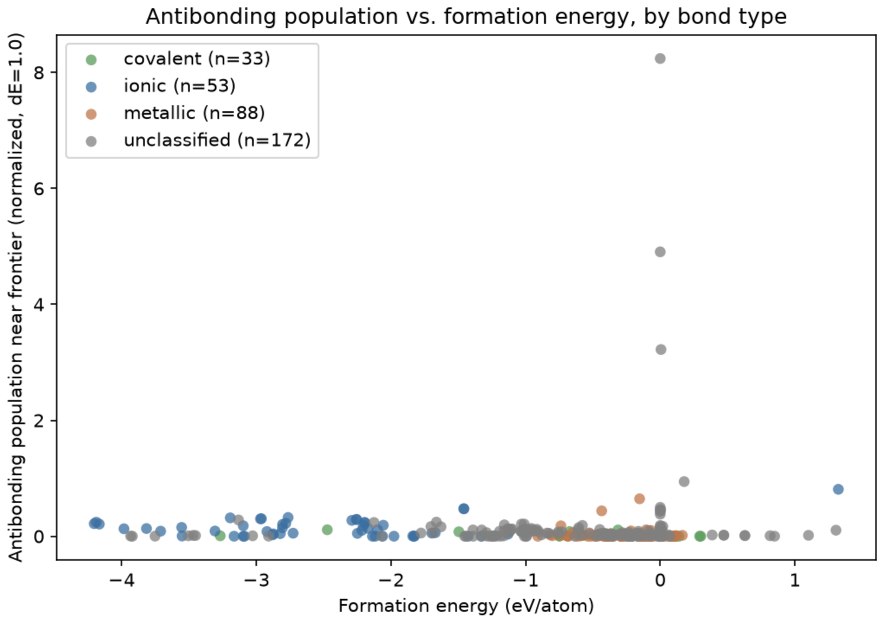
The Mn_2O_7 case is the manuscript's own guard-rail example: exobondic by this static sign criterion, yet Mn_2O_7 is a real, observed compound that simply decomposes slowly (gradual O_2 loss) rather than by sudden bond rupture — `classify.py` structurally attaches a warning to *every* `UNSTABLE_NONEXISTENT` verdict for exactly this reason, rather than hardcoding Mn_2O_7 as a special case.

The full list of every compound and element computed across the entire `extension` campaign — not just `extension4`'s 89, but all four batches combined (163 compounds total: elemental references, carbon allotropes, TiO2 high-pressure polymorphs, and every ionic/ covalent compound added along the way) — together with the case-1 decomposition reactions computed from them (their own ΔICOHP/atom and endobondic/exobondic label, distinct from the 7 literature cases above), is given in Appendix E through Appendix H.

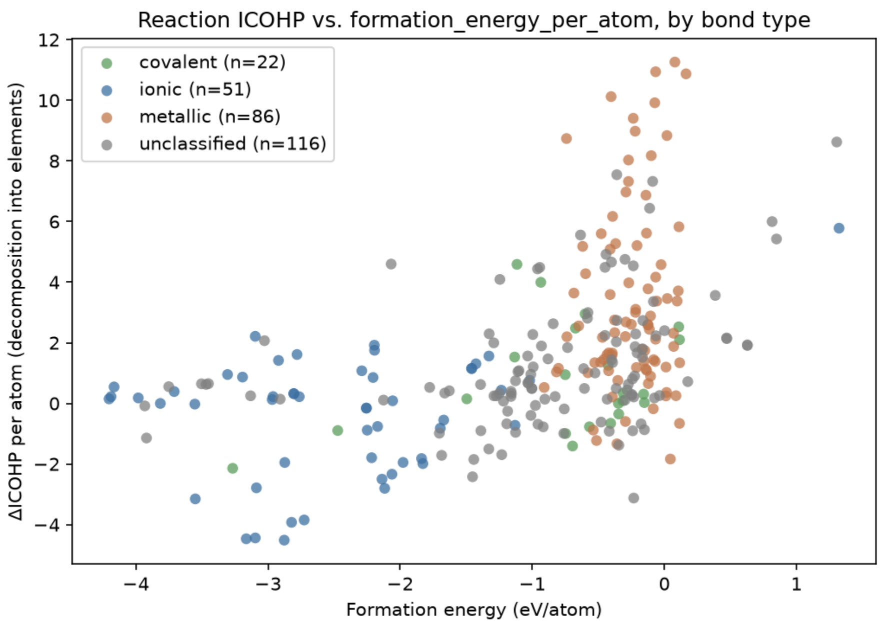
12.2 Missions #3–#5, reprocessed at 349 compounds

Every per-compound descriptor script was rerun unmodified over the grown dataset — `extension4`'s constituent elements were all already covered by the existing elemental-reference set, so no new element calculations were needed.

Mission #4 (antibonding): $\rho = -0.282$, $p = 9.2 \times 10^{-8}$ ($n = 346$) — weaker than the original $\rho = -0.328$, but the `bond_type=covalent` subgroup got *stronger* ($n = 33$, $\rho = -0.551$, $p = 0.0009$, vs. the original $\rho = -0.490$ at $n = 23$) and remains the single most robust stratified result in the project. `is_metal=False` newly survives ($n = 157$, $p = 0.001$). The old `bond_type=ionic` result ($n = 9$, $\rho = -0.683$) did not replicate once the sample grew to $n = 53$ ($\rho = -0.184$, not significant) — a clean demonstration of why $n < 15$ subgroups are flagged rather than trusted.



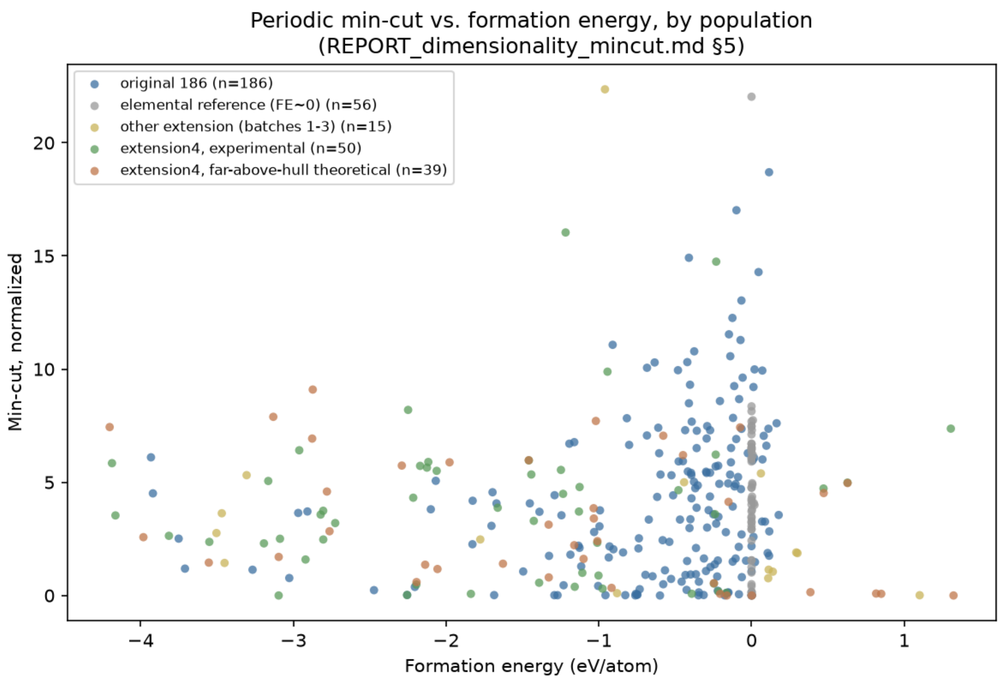
Mission #5 (reaction ICOHP): $\rho = 0.502$, $p = 6.3 \times 10^{-19}$ ($n = 275$) — up from $\rho = 0.370$, and now the clear strongest correlation in the project by a wider margin than before. `bond_type` stratification still fully explains that layer of the pooled signal (Kruskal–Wallis $p = 1.1 \times 10^{-15}$, no stratum survives) — **but `is_metal` stratification now survives on both sides** (`is_metal=False`: $\rho = 0.319$, $p = 0.0001$; `is_metal=True`: $\rho = 0.256$, $p = 0.0025$), where at $n = 186$ neither did. This revises the previous verdict that "the between-group confound explains the entirety of the observable signal" — it explains the `bond_type`-level pooling completely, but not a real, newly-detectable within-`is_metal`-group relationship. Case 2 (polymorph comparison), extended from 8 to **49 groups** specifically because `extension4` was designed to fix that sampling gap: 53.1% agreement between most-bonding and most-stable member, an exact Poisson-binomial test against the 45.9% chance baseline gives $P = 0.19$ — still no signal, but now a real test instead of an 8-point exploratory pass.



12.3 Why did the min-cut correlation collapse?

Mission #3's headline, $\rho = 0.285$ at $n = 186$, collapsed to $\rho = 0.089$, $p = 0.098$ (not significant) at $n = 349$. **This is not a refutation of the original result** — isolated, the original 186 compounds still give $\rho = 0.2845$, $p = 8.3 \times 10^{-5}$, essentially identical to the mission-#3 number. The collapse is a dilution artifact from pooling populations added for unrelated reasons:

- 65 of the 74 extension-batch-1–3 compounds are single-element reference structures (added for mission #5's elemental references), sitting at `formation_energy_per_atom`≈ 0 by construction with widely-scattered min-cut values — mechanically dilutes a rank correlation regardless of any true relationship. Excluding them recovers $\rho = 0.2157$ ($n = 193$).
- `extension4` is not one population but two, with *opposite-sign* correlations: the experimental half is flat ($\rho = 0.06$, not significant, $n = 50$); the deliberately far-above-hull theoretical half shows a *significant* correlation of the *opposite* sign ($\rho = -0.40$, $p = 0.011$, $n = 39$) — not established here whether this reflects real physics of far-from-hull structures or a selection artifact of picking the single highest-EAH entry per formula.



Recommendation: any future min-cut-vs-formation-energy number (or, by the same logic, any other descriptor's) should be conditioned on `family`/selection-provenance, the same way this project already conditions on `bond_type`/`is_metal` — pooling compounds selected for unrelated reasons is exactly the between-group confound this project's own methodology exists to catch, and it had not been applied to the dataset's own growing denominator until now.

Still no SISSO. Reaction ICOHP is now the strongest raw correlate of `formation_energy_per_atom` in the project, and the first descriptor whose signal is not fully explained away by between-group clustering at every level tested — the strongest argument yet for a stratified (`is_metal`-conditional) feature search over a pooled one, still not attempted.

Detailed reports: `analysis/REPORT_dimensionality_mincut.md`, `analysis/REPORT_antibonding.md`, `analysis/REPORT_reaction_icohp.md`.

A Compound List

The 186 compounds in the dataset, grouped by category (family), with their Materials Project identifier, PBE thermodynamic hull distance (`energy_above_hull`), and the minimum ICOHP percolation weight (`icohp_percolation_weight_min`, this work's main metric). Auto-generated from `analysis/percolation_vs_hull.csv` by `report/gen_appendix.py`.

Table 1: Complete list of the 186 compounds, grouped by category, with their Materials Project identifier, PBE hull distance, and minimum ICOHP percolation weight.

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
Experimental, stable (hull) ($n = 63$)				
As2S3	mp-1189572	covalent	0.0000	1.00e-05
AsF3	mp-28027	covalent	0.0000	0.00e+00
BI3	mp-23189	covalent	0.0000	1.00e-05
BW	mp-7832	–	0.0000	1.90e-04
BaAu2	mp-30363	metallic	0.0000	5.19e-02
BaCl2	mp-568662	ionic	0.0000	1.31e-02
BaH2	mp-23715	–	0.0000	2.40e-04
Be2Cu	mp-2031	metallic	0.0000	1.15e-03
Be2Nb3	mp-11272	metallic	0.0000	6.36e-03
Be2V	mp-11281	metallic	0.0000	6.80e-04
CaAl4	mp-1749	metallic	0.0000	1.03e-02
CaSn	mp-2450	metallic	0.0000	5.23e-03
CdPd	mp-1696	metallic	0.0000	9.00e-05
CoB	mp-20857	–	0.0000	6.00e-05
Cs2As3	mp-15557	–	0.0000	2.02e-02
FeS	mp-505531	–	0.0000	3.05e-03
Ge2Os	mp-10032	metallic	0.0000	1.21e-02
HfTc2	mp-1095669	metallic	0.0000	1.77e-02
Hg4Pt	mp-936	metallic	0.0000	3.36e-03
HgF2	mp-8177	–	0.0000	8.95e-03
InBr	mp-22870	covalent	0.0000	1.02e-02
KI	mp-22898	ionic	0.0000	2.26e-02
KN3	mp-827	–	0.0000	2.80e-04
Li2Se	mp-2286	–	0.0000	1.16e-02
LiB	mp-1001835	–	0.0000	2.91e-03
Mg2Ni	mp-2137	metallic	0.0000	1.60e-04
MgP4	mp-384	–	0.0000	0.00e+00
MnAl12	mp-1104165	metallic	0.0000	1.42e-02
MoSe2	mp-1634	covalent	0.0000	1.16e-03
NaS	mp-2400	–	0.0000	0.00e+00
Na	mp-10172	–	0.0000	1.97e-03
Nb3Ir	mp-1458	metallic	0.0000	9.02e-03
NbGa3	mp-1973	metallic	0.0000	8.58e-03
PI2	mp-29443	covalent	0.0000	2.30e-04
PbSe	mp-2201	covalent	0.0000	1.66e-02
RbI	mp-22903	ionic	0.0000	2.00e-02
Sc5Ge3	mp-17190	metallic	0.0000	5.30e-04
ScAg2	mp-1018128	metallic	0.0000	1.70e-04
ScTe	mp-10026	–	0.0000	7.12e-03
SnPt	mp-19856	metallic	0.0000	1.29e-02
SrO2	mp-2697	ionic	0.0000	4.40e-04
SrSi	mp-2661	metallic	0.0000	7.00e-05
Ta5Ge3	mp-17593	metallic	0.0000	3.46e-02

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
TaP	mp-1067587	–	0.0000	3.14e-02
TaSb2	mp-11697	metallic	0.0000	1.17e-02
Te2Pd	mp-782	–	0.0000	1.44e-02
Te2Ru	mp-267	covalent	0.0000	4.34e-03
TeO2	mp-8377	covalent	0.0000	4.00e-05
Ti3In	mp-866046	metallic	0.0000	5.01e-03
Ti3Pt	mp-2339	metallic	0.0000	1.78e-03
TiBe12	mp-1104067	metallic	0.0000	2.40e-04
TiNi	mp-1048	metallic	0.0000	3.91e-03
TiO	mp-1071163	–	0.0000	6.40e-04
TiSi2	mp-1077503	metallic	0.0000	6.18e-03
TlCl	mp-571079	–	0.0000	7.17e-03
VNi2	mp-11531	metallic	0.0000	3.88e-03
WO2	mp-1524394	–	0.0000	2.20e-04
Zr2Au	mp-2348	metallic	0.0000	3.06e-02
Zr5Pb3	mp-681992	metallic	0.0000	2.00e-05
ZrMo2	mp-2049	metallic	0.0000	1.05e-02
Si	mp-149	covalent	0.0000	3.17e-03
NaCl	mp-22862	ionic	0.0000	3.07e-02
AlNi	mp-1487	metallic	0.0000	4.19e-03
Experimental, metastable ($n = 63$)				
CdI2	mp-567259	–	0.0005	2.38e-02
MgCl2	mp-570259	ionic	0.0006	1.90e-04
C	mp-169	covalent	0.0008	2.50e-04
ZrBr	mp-1065846	–	0.0008	7.88e-03
Li3Hg	mp-1646	metallic	0.0013	1.20e-03
Ta7Co6	mp-1104548	metallic	0.0015	3.10e-04
CaGa4	mp-2461	metallic	0.0016	1.91e-02
InCl	mp-571636	covalent	0.0016	8.50e-03
NiB	mp-14019	–	0.0018	4.10e-04
MgH2	mp-23711	–	0.0019	1.00e-05
Sn7Ir5	mp-20466	metallic	0.0024	1.00e-02
NaN3	mp-1066400	–	0.0029	7.00e-04
ClF3	mp-556767	–	0.0030	0.00e+00
SiS2	mp-1095294	covalent	0.0030	1.00e-04
Sr5Sn3	mp-17720	metallic	0.0030	1.08e-02
Te2Ir	mp-1551	–	0.0030	2.80e-03
Li13Sn5	mp-30769	metallic	0.0033	8.50e-04
H3N	mp-643432	covalent	0.0038	0.00e+00
Cs2Se	mp-1011696	–	0.0042	2.62e-02
In3Ir	mp-630976	metallic	0.0043	2.04e-03
HI	mp-697025	–	0.0044	0.00e+00
Rb2Hg7	mp-31474	metallic	0.0045	3.14e-02
PbAu2	mp-19871	metallic	0.0047	2.01e-03
BeCu	mp-2323	metallic	0.0050	4.80e-04
SiO2	mp-546794	covalent	0.0056	1.50e-04
IF7	mp-27988	–	0.0063	0.00e+00
K	mp-10157	–	0.0069	8.07e-02
Al3Os2	mp-16521	metallic	0.0074	6.23e-03
Nb2O5	mp-1595	–	0.0075	1.00e-04
LiBr	mp-23259	ionic	0.0099	9.88e-02
ZnAg	mp-1912	metallic	0.0112	9.67e-03
MgPd3	mp-7753	metallic	0.0114	2.21e-03
AsPd2	mp-1080831	–	0.0118	3.50e-04

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
GeSe2	mp-10074	covalent	0.0125	4.00e-05
Re3Ge7	mp-29141	metallic	0.0126	2.42e-03
Cu3Ge	mp-19724	metallic	0.0126	5.30e-04
Ti3Ge	mp-1189044	metallic	0.0130	1.40e-02
ZrGa3	mp-30685	metallic	0.0130	7.64e-03
SrAl2	mp-1638	metallic	0.0136	5.36e-03
NiS	mp-1547	–	0.0142	3.00e-04
Al12W	mp-11227	metallic	0.0152	1.37e-02
CsH	mp-632319	–	0.0161	2.66e-02
CaGa2	mp-992	metallic	0.0170	1.66e-02
NiHg4	mp-570835	metallic	0.0171	1.09e-03
SiRh	mp-2287	metallic	0.0246	1.10e-02
BaSn5	mp-571353	metallic	0.0262	4.58e-03
PS	mp-612	covalent	0.0265	8.00e-05
AsRh	mp-22079	–	0.0269	1.93e-03
Te2Au	mp-567525	–	0.0274	1.62e-02
Be3N2	mp-6977	–	0.0315	2.00e-05
Y2O3	mp-558573	–	0.0342	5.00e-05
AgO	mp-1079720	–	0.0423	4.00e-05
FeTe2	mp-1102002	–	0.0449	5.20e-04
H2O	mp-2739191	–	0.0494	0.00e+00
CdO2	mp-2310	–	0.0562	9.00e-05
Mo3Se4	mp-21021	–	0.0609	1.84e-03
PbSe	mp-22009	covalent	0.0731	1.68e-02
Be12Pd	mp-12666	metallic	0.0795	5.00e-05
Sr2As	mp-15697	–	0.0800	2.06e-03
AsRh2	mp-1102364	–	0.0806	4.63e-03
ZrRh	mp-2808	metallic	0.0839	3.50e-02
K2S	mp-559278	–	0.0873	1.07e-02
Ge4Rh	mp-1104286	metallic	0.0907	2.08e-03
Theoretical, metastable ($n = 60$)				
Li3Ag	mp-976408	metallic	0.0013	1.15e-02
Y3Co	mp-1189329	metallic	0.0016	1.50e-02
TiNi	mp-1067248	metallic	0.0020	5.29e-03
TaP	mp-1187244	–	0.0033	3.59e-02
Sr3As4	mp-678007	–	0.0062	4.47e-03
Ga2Se3	mp-1224809	covalent	0.0063	1.14e-02
MnN	mp-1009130	covalent	0.0151	1.85e-03
BeCl2	mp-1190080	ionic	0.0187	2.37e-03
MgSn	mp-1094801	metallic	0.0212	4.26e-03
TiW	mp-1216621	metallic	0.0225	2.25e-03
SrTl3	mp-1206636	metallic	0.0294	3.29e-03
IrCl3	mp-729507	–	0.0301	2.72e-03
Ta4C3	mp-1218000	–	0.0335	1.44e-03
Sc2O3	mp-755313	–	0.0364	5.10e-04
SnBr2	mp-978985	covalent	0.0405	3.40e-04
Sn7Pd5	mp-1219006	metallic	0.0414	2.30e-03
Tl4Sn	mp-1216560	metallic	0.0459	1.00e+00
LiTl3	mp-1185253	metallic	0.0467	3.35e-03
PtCl2	mp-1219748	–	0.0490	5.04e-03
HCl	mp-684609	–	0.0536	2.50e-04
MgSn	mp-1094669	metallic	0.0560	2.60e-04
K5As4	mp-684799	–	0.0564	8.80e-03
KSn3	mp-1185151	metallic	0.0573	2.65e-02

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
NbS2	mp-774873	–	0.0576	1.20e-04
YAg3	mp-1187811	metallic	0.0594	1.12e-03
B3Ir2	mp-1228647	–	0.0609	2.80e-04
TiMo2	mp-1216675	metallic	0.0613	3.77e-02
Cr2C	mp-1226378	–	0.0625	2.08e-03
ZrO2	mp-775909	–	0.0627	1.20e-04
Zn	mp-2647117	–	0.0717	1.04e-02
IN2	mp-1097011	–	0.0730	4.70e-04
CoSe2	mp-1390196	–	0.0734	8.40e-04
GeS	mp-12910	covalent	0.0762	2.26e-03
ZrTi	mp-1215236	metallic	0.0883	3.15e-02
VCo3	mp-1095057	metallic	0.0942	2.93e-03
Tl3Ga	mp-1187539	metallic	0.0972	5.00e-05
CaGe2	mp-643542	metallic	0.0979	1.30e-02
NbRh2	mp-1220414	metallic	0.1043	1.76e-02
TcW	mp-1217337	metallic	0.1056	4.10e-03
MgF2	mp-560236	ionic	0.1076	2.00e-05
VN	mp-1017532	–	0.1080	4.00e-05
Na2Cl	mp-990084	–	0.1105	3.76e-02
ZnPb3	mp-1187949	metallic	0.1107	4.51e-03
Sr3Sn	mp-1187136	metallic	0.1111	7.94e-03
TaRe	mp-1217894	metallic	0.1113	6.98e-02
Al4Si	mp-1228120	metallic	0.1149	8.54e-02
B6Pb	mp-1096825	covalent	0.1151	2.35e-03
ScTc3	mp-867262	metallic	0.1154	3.31e-02
CrMo	mp-1226200	metallic	0.1163	4.23e-02
BPd5	mp-1228665	–	0.1325	2.42e-02
Zr3Be	mp-1188016	metallic	0.1430	1.05e-03
Na3Cl	mp-1064484	–	0.1456	2.55e-03
SiAu3	mp-1186998	metallic	0.1643	5.68e-03
InSe2	mp-1232036	–	0.1733	1.64e-03
TiSi2	mp-570991	metallic	0.1796	1.35e-02
BeAu3	mp-983539	metallic	0.1818	2.41e-02
Zr3Ge	mp-1188039	metallic	0.1851	6.59e-03
AlSb	mp-1023918	metallic	0.1878	3.60e-04
Mg15B	mp-1023515	–	0.1943	2.80e-04
HfMo	mp-1224314	metallic	0.1991	2.00e-02

B VASP Calculation Parameters

Fixed INCAR parameters, identical across all 186 compounds (static, LOBSTER-compatible run) — only `NBANDS` (appendix D) and the k-point grid vary by compound.

Parameter	Value	Description
`PREC`	Accurate	Overall precision (charge grid, projectors)
`ENCUT`	600.0	Plane-wave cutoff energy (eV)
`EDIFF`	1e-06	Electronic convergence criterion (eV)
`IBRION`	-1	Ionic dynamics type (-1 = static, no relaxation)
`NSW`	0	Number of ionic steps (0 = static run on the MP structure)
`ISMEAR`	0	Smearing type (0 = Gaussian)
`SIGMA`	0.05	Smearing width (eV)
`LASPH`	True	Non-spherical corrections to the PAW potential
`ISYM`	-1	Symmetry (-1 = disabled, required by LOBSTER)
`LWAVE`	True	Write WAVECAR (required by LOBSTER)
`LCHARG`	True	Write CHGCAR
`NPAR`	4	Band-level parallelization
`KPAR`	2	K-point-level parallelization

C PAW-PBE Potentials

One PAW-PBE potential per element present in the dataset (pymatgen's `MPRelaxSet` library — the same one Materials Project used to relax the input structures), with one documented exception for tungsten (§7): the recommended `W_pv` variant is absent from the local mirror, and `W_sv` turned out to be corrupted (missing `LEXCH` field) — plain `W` is used instead, verified valid.

Table 2: PAW-PBE potentials used, one per element present in the dataset (pymatgen's MPRelaxSet library, with one documented exception: W substituted for W_pv/W_sv, see §7).

Element	PAW-PBE potential
Ag	`Ag (02Apr2005)`
Al	`Al (04Jan2001)`
As	`As (22Sep2009)`
Au	`Au (04Oct2007)`
B	`B (06Sep2000)`
Ba	`Ba_sv (06Sep2000)`
Be	`Be_sv (06Sep2000)`
Br	`Br (06Sep2000)`
C	`C (08Apr2002)`
Ca	`Ca_sv (06Sep2000)`
Cd	`Cd (06Sep2000)`
Cl	`Cl (06Sep2000)`
Co	`Co (02Aug2007)`
Cr	`Cr_pv (02Aug2007)`
Cs	`Cs_sv (25Jan2019)`
Cu	`Cu_pv (06Sep2000)`
F	`F (08Apr2002)`
Fe	`Fe_pv (02Aug2007)`
Ga	`Ga_d (06Jul2010)`
Ge	`Ge_d (03Jul2007)`
H	`H (15Jun2001)`
Hf	`Hf_pv (06Sep2000)`

Element	PAW-PBE potential
Hg	`Hg (06Sep2000)`
I	`I (08Apr2002)`
In	`In_d (06Sep2000)`
Ir	`Ir (06Sep2000)`
K	`K_sv (06Sep2000)`
Li	`Li_sv (10Sep2004)`
Mg	`Mg_pv (13Apr2007)`
Mn	`Mn_pv (02Aug2007)`
Mo	`Mo_pv (04Feb2005)`
N	`N (08Apr2002)`
Na	`Na_pv (19Sep2006)`
Nb	`Nb_pv (08Apr2002)`
Ni	`Ni_pv (06Sep2000)`
O	`O (08Apr2002)`
Os	`Os_pv (20Jan2003)`
P	`P (06Sep2000)`
Pb	`Pb_d (06Sep2000)`
Pd	`Pd (04Jan2005)`
Pt	`Pt (04Feb2005)`
Rb	`Rb_sv (06Sep2000)`
Re	`Re_pv (06Sep2000)`
Rh	`Rh_pv (25Jan2005)`
Ru	`Ru_pv (28Jan2005)`
S	`S (06Sep2000)`
Sb	`Sb (06Sep2000)`
Sc	`Sc_sv (07Sep2000)`
Se	`Se (06Sep2000)`
Si	`Si (05Jan2001)`
Sn	`Sn_d (06Sep2000)`
Sr	`Sr_sv (07Sep2000)`
Ta	`Ta_pv (07Sep2000)`
Tc	`Tc_pv (04Feb2005)`
Te	`Te (08Apr2002)`
Ti	`Ti_pv (07Sep2000)`
Tl	`Tl_d (06Sep2000)`
V	`V_pv (07Sep2000)`
W	`W (08Apr2002)`
Y	`Y_sv (25May2007)`
Zn	`Zn (06Sep2000)`
Zr	`Zr_sv (04Jan2005)`

D k-point Grid and NBANDS per Compound

k-point grid auto-generated by `pymatgen.io.vasp.Kpoints.automatic_density_by_vol` (density `kppvol=100`, Γ-centered or Monkhorst–Pack mesh depending on cell symmetry) and `NBANDS`, derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`) — a strictly sufficient value for the local-basis projection, not a fixed guess.

Table 3: k-point grid (`pymatgen` automatic density, `kppvol=100`) and number of bands (derived from the LOBSTER basis) per compound.

Formula	mp-id	k-point grid	NBANDS
Experimental, stable (hull) ($n = 63$)			
As2S3	mp-1189572	5 3 2 (G)	80
AsF3	mp-28027	5 4 4 (G)	64
BI3	mp-23189	4 4 4 (G)	32
BW	mp-7832	9 9 3 (G)	52
BaAu2	mp-30363	6 6 7 (G)	23
BaCl2	mp-568662	6 6 6 (G)	13
BaH2	mp-23715	6 4 3 (G)	28
Be2Cu	mp-2031	7 7 7 (G)	38
Be2Nb3	mp-11272	8 4 4 (M)	74
Be2V	mp-11281	7 7 4 (G)	76
CaAl4	mp-1749	7 7 4 (G)	21
CaSn	mp-2450	6 6 4 (M)	28
CdPd	mp-1696	7 6 6 (G)	36
CoB	mp-20857	9 7 5 (G)	52
Cs2As3	mp-15557	4 4 3 (G)	44
FeS	mp-505531	8 8 5 (G)	26
Ge2Os	mp-10032	9 6 4 (G)	54
HfTc2	mp-1095669	5 5 3 (G)	108
Hg4Pt	mp-936	5 5 5 (G)	45
HgF2	mp-8177	8 8 8 (G)	17
InBr	mp-22870	6 6 4 (M)	26
KI	mp-22898	6 6 6 (G)	9
KN3	mp-827	5 5 5 (G)	34
Li2Se	mp-2286	7 7 7 (G)	14
LiB	mp-1001835	7 7 9 (G)	18
Mg2Ni	mp-2137	5 5 2 (G)	102
MgP4	mp-384	5 5 3 (G)	40
MnAl12	mp-1104165	4 4 4 (M)	57
MoSe2	mp-1634	9 9 2 (G)	34
NaS	mp-2400	6 6 3 (G)	32
Na	mp-10172	8 8 4 (G)	8
Nb3Ir	mp-1458	5 5 5 (G)	72
NbGa3	mp-1973	8 8 6 (M)	36
PI2	mp-29443	6 4 3 (G)	24
PbSe	mp-2201	7 7 7 (G)	13
RbI	mp-22903	6 6 6 (G)	9
Sc5Ge3	mp-17190	3 3 5 (G)	154
ScAg2	mp-1018128	8 8 5 (G)	28
ScTe	mp-10026	7 7 4 (G)	28
SnPt	mp-19856	7 7 5 (G)	36
SrO2	mp-2697	8 8 7 (G)	13
SrSi	mp-2661	7 6 4 (G)	18
Ta5Ge3	mp-17593	4 4 4 (M)	144
TaP	mp-1067587	9 9 4 (G)	26
TaSb2	mp-11697	8 5 4 (G)	34
Te2Pd	mp-782	7 7 5 (G)	17
Te2Ru	mp-267	7 5 4 (G)	34
TeO2	mp-8377	6 5 3 (G)	48
Ti3In	mp-866046	5 5 6 (G)	72
Ti3Pt	mp-2339	5 5 5 (G)	72
TiBe12	mp-1104067	7 5 5 (G)	69

Table 3 (continued)

Formula	mp-id	k-point grid	NBANDS
TiNi	mp-1048	10 7 6 (G)	36
TiO	mp-1071163	10 6 6 (G)	39
TiSi2	mp-1077503	8 8 4 (M)	34
TlCl	mp-571079	6 6 4 (M)	26
VNi2	mp-11531	12 8 7 (G)	27
WO2	mp-1524394	6 5 5 (G)	68
Zr2Au	mp-2348	9 9 4 (G)	29
Zr5Pb3	mp-681992	3 3 5 (G)	154
ZrMo2	mp-2049	6 6 6 (G)	56
Si	mp-149	8 8 8 (G)	100
NaCl	mp-22862	8 8 8 (G)	100
AlNi	mp-1487	10 10 10 (M)	100
Experimental, metastable ($n = 63$)			
CdI2	mp-567259	7 7 4 (G)	17
MgCl2	mp-570259	8 8 5 (G)	12
C	mp-169	12 12 8 (M)	100
ZrBr	mp-1065846	8 8 3 (G)	28
Li3Hg	mp-1646	7 7 7 (G)	24
Ta7Co6	mp-1104548	6 6 3 (G)	117
CaGa4	mp-2461	7 7 5 (G)	41
InCl	mp-571636	6 6 4 (M)	26
NiB	mp-14019	10 10 7 (G)	26
MgH2	mp-23711	6 5 5 (G)	24
Sn7Ir5	mp-20466	4 4 4 (M)	108
NaN3	mp-1066400	7 7 7 (G)	16
ClF3	mp-556767	6 4 3 (G)	64
SiS2	mp-1095294	5 4 3 (G)	48
Sr5Sn3	mp-17720	3 3 3 (G)	104
Te2Ir	mp-1551	7 5 2 (G)	51
Li13Sn5	mp-30769	6 6 1 (G)	110
H3N	mp-643432	8 5 5 (G)	28
Cs2Se	mp-1011696	5 3 2 (G)	56
In3Ir	mp-630976	4 4 4 (M)	144
HI	mp-697025	3 3 3 (G)	40
Rb2Hg7	mp-31474	4 4 4 (G)	73
PbAu2	mp-19871	5 5 5 (G)	54
BeCu	mp-2323	10 10 10 (M)	100
SiO2	mp-546794	6 6 6 (M)	24
IF7	mp-27988	5 5 3 (G)	64
K	mp-10157	6 6 6 (G)	5
Al3Os2	mp-16521	9 9 3 (G)	30
Nb2O5	mp-1595	7 7 4 (G)	38
LiBr	mp-23259	8 8 8 (G)	100
ZnAg	mp-1912	9 9 9 (G)	18
MgPd3	mp-7753	7 7 7 (G)	31
AsPd2	mp-1080831	8 4 4 (G)	66
GeSe2	mp-10074	5 5 5 (G)	34
Re3Ge7	mp-29141	9 6 1 (G)	180
Cu3Ge	mp-19724	6 6 5 (G)	72
Ti3Ge	mp-1189044	4 4 4 (M)	144
ZrGa3	mp-30685	7 7 5 (G)	37
SrAl2	mp-1638	5 5 5 (G)	26
NiS	mp-1547	6 6 6 (M)	39
Al12W	mp-11227	4 4 4 (M)	57

Table 3 (continued)

Formula	mp-id	k-point grid	NBANDS
CsH	mp-632319	7 7 7 (G)	6
CaGa2	mp-992	7 7 6 (G)	23
NiHg4	mp-570835	6 6 6 (M)	45
SiRh	mp-2287	6 6 5 (G)	52
BaSn5	mp-571353	5 5 4 (G)	50
PS	mp-612	4 4 3 (G)	64
AsRh	mp-22079	8 5 4 (G)	52
Te2Au	mp-567525	7 7 5 (G)	17
Be3N2	mp-6977	10 10 3 (G)	46
Y2O3	mp-558573	8 4 3 (G)	96
AgO	mp-1079720	8 5 5 (G)	52
FeTe2	mp-1102002	4 4 4 (M)	68
H2O	mp-2739191	7 7 4 (G)	24
CdO2	mp-2310	5 5 5 (G)	68
Mo3Se4	mp-21021	4 4 4 (M)	86
PbSe	mp-22009	6 6 2 (G)	26
Be12Pd	mp-12666	6 6 6 (M)	69
Sr2As	mp-15697	6 6 3 (G)	28
AsRh2	mp-1102364	7 4 3 (G)	88
ZrRh	mp-2808	8 8 8 (M)	19
K2S	mp-559278	5 5 4 (G)	28
Ge4Rh	mp-1104286	4 4 3 (G)	135
Theoretical, metastable ($n = 60$)			
Li3Ag	mp-976408	7 7 7 (G)	24
Y3Co	mp-1189329	3 4 4 (G)	156
TiNi	mp-1067248	10 7 6 (G)	36
TaP	mp-1187244	9 9 9 (G)	13
Sr3As4	mp-678007	3 3 4 (G)	62
Ga2Se3	mp-1224809	5 5 5 (G)	30
MnN	mp-1009130	10 10 10 (G)	13
BeCl2	mp-1190080	3 3 3 (G)	78
MgSn	mp-1094801	8 8 6 (M)	13
TiW	mp-1216621	10 10 6 (M)	18
SrTl3	mp-1206636	5 5 5 (G)	32
IrCl3	mp-729507	3 3 6 (G)	84
Ta4C3	mp-1218000	6 6 6 (M)	48
Sc2O3	mp-755313	6 5 6 (G)	64
SnBr2	mp-978985	4 4 6 (M)	34
Sn7Pd5	mp-1219006	5 5 3 (G)	108
Tl4Sn	mp-1216560	5 5 5 (G)	45
LiTl3	mp-1185253	6 6 6 (M)	32
PtCl2	mp-1219748	5 5 4 (G)	34
HCl	mp-684609	7 7 7 (G)	5
MgSn	mp-1094669	8 8 8 (M)	13
K5As4	mp-684799	4 4 3 (G)	41
KSn3	mp-1185151	4 4 5 (G)	64
NbS2	mp-774873	10 6 2 (M)	34
YAg3	mp-1187811	6 6 6 (G)	37
B3Ir2	mp-1228647	9 5 4 (G)	60
TiMo2	mp-1216675	5 6 14 (G)	27
Cr2C	mp-1226378	10 10 7 (G)	22
ZrO2	mp-775909	10 6 4 (M)	36
Zn	mp-2647117	11 11 11 (G)	9
IN2	mp-1097011	5 5 5 (G)	24

Table 3 (continued)

Formula	mp-id	k-point grid	NBANDS
CoSe2	mp-1390196	4 4 4 (G)	68
GeS	mp-12910	7 7 5 (G)	26
ZrTi	mp-1215236	9 9 6 (G)	19
VCo3	mp-1095057	7 6 6 (G)	72
Tl3Ga	mp-1187539	5 5 5 (G)	36
CaGe2	mp-643542	7 7 6 (G)	23
NbRh2	mp-1220414	10 6 4 (M)	54
TcW	mp-1217337	10 10 6 (M)	18
MgF2	mp-560236	6 6 5 (G)	24
VN	mp-1017532	11 11 5 (G)	26
Na2Cl	mp-990084	6 6 6 (M)	12
ZnPb3	mp-1187949	5 5 5 (G)	36
Sr3Sn	mp-1187136	5 5 5 (G)	24
TaRe	mp-1217894	10 10 6 (M)	18
Al4Si	mp-1228120	6 6 6 (M)	20
B6Pb	mp-1096825	6 6 6 (M)	33
ScTc3	mp-867262	7 7 7 (G)	37
CrMo	mp-1226200	11 11 7 (G)	18
BPd5	mp-1228665	6 6 6 (M)	49
Zr3Be	mp-1188016	6 6 6 (M)	35
Na3Cl	mp-1064484	8 8 3 (G)	16
SiAu3	mp-1186998	7 7 7 (G)	31
InSe2	mp-1232036	7 7 3 (G)	34
TiSi2	mp-570991	8 8 8 (M)	17
BeAu3	mp-983539	7 7 7 (G)	32
Zr3Ge	mp-1188039	4 4 6 (G)	78
AlSb	mp-1023918	6 6 5 (G)	16
Mg15B	mp-1023515	4 4 3 (G)	64
HfMo	mp-1224314	10 10 6 (M)	18

E Extension Campaign: Elements and Allotropes

Every single-element structure computed across all four `extension` batches (§12; `download_extension.py` through `download_extension4.py`): the 62-element decomposition reference set (`ELEMENT_REFERENCE`) plus the additional carbon allotropes (diamond, lonsdaleite, M-carbon, W-carbon) computed alongside the graphite reference specifically to test descriptors against structures whose true stable regime is elsewhere. An experimental compound's formula is marked with a superscript star; unmarked formulas are theoretical. Auto-generated from every `mp_metadata.json` with `family=="extension"` by `report/gen_appendix_extension.py`.

Table 4: Elements and allotropes of the `extension` campaign (batches 1–4): references used to decompose multi-element compounds, plus additional carbon allotropes (TiO2's polymorphs are a binary compound, see the compounds table instead). * = experimental reference (unmarked = theoretical). No ΔICOHP column: an element is not itself a decomposition reaction.

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
Ag*	mp-124	Fm-3m	0.0021
Al*	mp-134	Fm-3m	0.0000
As*	mp-158	Cmce	0.0000
Au*	mp-81	Fm-3m	0.0000

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
B*	mp-160	R-3m	0.0000
Ba*	mp-122	Im-3m	0.0000
Be*	mp-87	P6_3/mmc	0.0000
Br*	mp-23154	Cmce	0.0000
C	mp-1080826	C2/m	0.3009
C	mp-1190171	Pnma	0.2927
C*	mp-66	Fd-3m	0.1123
C*	mp-48	P6_3/mmc	0.0031
C*	mp-47	P6_3/mmc	0.1395
Ca*	mp-21	Im-3m	0.0056
Cd*	mp-94	P6_3/mmc	0.0000
Cl2*	mp-22848	Cmce	0.0000
Co*	mp-102	Fm-3m	0.0000
Cr*	mp-90	Im-3m	0.0000
Cs*	mp-1	Im-3m	0.0193
Cu*	mp-30	Fm-3m	0.0000
F2*	mp-561203	C2/c	0.0031
Fe*	mp-13	Im-3m	0.0000
Ga*	mp-142	Cmce	0.0000
Ge*	mp-32	Fd-3m	0.0000
H2*	mp-730101	P2_12_12_1	0.0000
Hf*	mp-103	P6_3/mmc	0.0000
Hg*	mp-10861	P6/mmm	0.0030
I*	mp-23153	Cmce	0.0000
In*	mp-1055994	I4/mmm	0.0045
Ir*	mp-101	Fm-3m	0.0000
Li*	mp-1018134	R-3m	0.0000
Mg*	mp-153	P6_3/mmc	0.0000
Mn*	mp-35	I-43m	0.0000
Mo*	mp-129	Im-3m	0.0000
N2	mp-1059834	Imma	1.1010
Nb*	mp-75	Im-3m	0.0000
Ni*	mp-23	Fm-3m	0.0000
O2*	mp-1524462	C2/m	0.0000
Os*	mp-49	P6_3/mmc	0.0000
P*	mp-568348	P2/c	0.0000
Pb*	mp-20483	Fm-3m	0.0000
Pd*	mp-2	Fm-3m	0.0000
Pt*	mp-126	Fm-3m	0.0000
Rb*	mp-70	Im-3m	0.0092
Re*	mp-8	P6_3/mmc	0.0036
Rh*	mp-74	Fm-3m	0.0000
Ru*	mp-33	P6_3/mmc	0.0000
S*	mp-77	Fddd	0.0000
Sb*	mp-104	R-3m	0.0000
Sc*	mp-67	P6_3/mmc	0.0000
Se*	mp-14	P3_121	0.0011
Sn*	mp-623511	I4/mmm	0.0614
Sr*	mp-139	P6_3/mmc	0.0000
Ta*	mp-50	Im-3m	0.0091
Tc*	mp-113	P6_3/mmc	0.0000
Te*	mp-19	P3_121	0.0000
Ti*	mp-46	P6_3/mmc	0.0152
Tl*	mp-82	P6_3/mmc	0.0000
V*	mp-146	Im-3m	0.0000
W*	mp-91	Im-3m	0.0000

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
Y*	mp-112	P6_3/mmc	0.0000
Zn*	mp-79	P6_3/mmc	0.0000
Zr*	mp-131	P6_3/mmc	0.0000

F Extension Campaign: Compounds and Polymorphs

Every multi-element target compound across all four `extension` batches (100 total, 89 from `extension4`) — including the TiO2 high-pressure polymorphs (`extension2`) alongside every `extension4` alkali/alkaline-earth binary — with ΔICOHP per atom and its endobondic/exobondic label where a case-1 decomposition reaction could be computed. Auto-generated from `analysis/reaction_icoh` and `analysis/reaction_analysis_case1_full.csv` by `report/gen_appendix_extension.py`.

Table 5: Target compounds and polymorphs of the `extension` campaign (batches 1–4, 163 compounds, 89 from `extension4`) – includes TiO2's high-pressure polymorphs and every multi-element compound from earlier batches. * = experimental compound (unmarked = theoretical). ΔICOHP per atom, products − reactants convention (`reaction_analysis`); "–" where no case-1 reaction could be computed (missing elemental reference).

Formula	mp-id / COD	Space group	E_{hull} (eV/at)	ΔICOHP/at. (eV)	Label
Ba5P4*	mp-17566	Pnma	0.0000	-0.9590	exobondic
BaCl2*	mp-23199	Pnma	0.0068	-0.2244	exobondic
BaF3*	mp-1080821	P2_1/m	0.0000	-0.8680	exobondic
BaF3	mp-1183303	I4/mmm	0.0612	-0.2493	exobondic
BaN2*	mp-1102371	Cc	2.0568	-8.6248	exobondic
BaO*	mp-1008500	P6_3/mmc	0.0189	-0.3221	exobondic
BaO	mp-1950989	P6_3/mmc	0.0434	-1.6129	exobondic
BaS3*	mp-556296	P2_12_12	0.0436	-0.6233	exobondic
Be3N2*	mp-18337	Ia-3	0.0000	-0.0746	exobondic
Be3N2	mp-1017550	P-3m1	0.2081	0.3960	endobondic
Be3P2*	mp-567841	Ia-3	0.0000	-0.2111	exobondic
Be3P2	mp-1084771	Pn-3m	0.1567	-0.1795	exobondic
BeCl2*	mp-23267	Ibam	0.0087	1.8204	endobondic
BeCl2	mp-1172909	I-43m	3.1688	-5.7807	exobondic
BeF2*	mp-15951	P3_121	0.0004	3.1514	endobondic
BeF2	mp-975644	P6/mmm	2.5181	-0.6816	exobondic
BeO*	mp-7599	P4_2/mnm	0.0153	2.7864	endobondic
BeO	mp-1778	F-43m	0.0070	4.4391	endobondic
Ca3N2*	mp-1047	R-3c	0.0167	-4.4881	exobondic
Ca3N2*	mp-844	Ia-3	0.0000	-4.4374	exobondic
Ca3N2	mp-1013524	Pm-3m	0.5112	-4.4898	exobondic
CaCl2*	mp-23214	Pnnm	0.0011	-0.3199	exobondic
CaCl2	mp-1120739	Fm-3m	0.0422	-0.2208	exobondic
CaF2*	mp-10464	Pnma	0.0346	-0.2227	exobondic
CaF2	mp-554355	P4/mmm	0.2403	-0.1814	exobondic
CaN*	mp-1058549	Fm-3m	0.3982	-4.9183	exobondic
CaO*	mp-1079707	Cmce	0.2098	-2.2127	exobondic
CaO*	mp-2605	Fm-3m	0.0000	-0.9525	exobondic
CaO	mp-1181903	Fd-3m	2.7281	-2.9971	exobondic
Cs2S*	mp-1077814	P6_3/mmc	0.0667	0.9554	endobondic
CsCl*	mp-573697	Fm-3m	0.0071	0.8830	endobondic
CsF*	mp-8455	Pm-3m	0.1088	3.8464	endobondic
CsN3*	mp-510557	I4/mcm	0.0000	-2.2903	exobondic

Formula	mp-id / COD	Space group	E_{hull} (eV/at)	ΔICOHP/at. (eV)	Label
CsN3	mp-1225892	P4/mmm	0.0113	-1.6664	exobondic
CsO2*	mp-1441	I4/mmm	0.0349	0.0666	endobondic
CsO2	mp-1096936	Cmmm	0.0940	0.7743	endobondic
CsP7*	mp-1201790	Pbcm	0.0000	-0.2978	exobondic
CsP7	mp-1213206	Pca2_1	0.0001	-0.2667	exobondic
K2O*	mp-971	Fm-3m	0.0000	-0.4357	exobondic
K2O	mp-1223784	P4/mmm	0.2000	-1.5638	exobondic
K2S*	mp-1022	Fm-3m	0.0000	-0.3359	exobondic
KCl*	mp-23289	Pm-3m	0.0776	0.7616	endobondic
KF3*	mp-1544557	P2_12_12_1	0.0125	-1.1511	exobondic
KF3	mp-2749823	P2_1	0.0119	-1.1505	exobondic
KN3	mp-636056	I4/mcm	1.0370	-5.9991	exobondic
KP*	mp-1058315	R-3m	1.0080	-2.1600	exobondic
KP	mp-1057787	Immm	1.0092	-2.1366	exobondic
Li2O*	mp-1960	Fm-3m	0.0000	2.3366	endobondic
Li2O	mp-755894	Pnma	0.0844	1.9481	endobondic
Li2S*	mp-1125	Pnma	0.0615	1.8505	endobondic
Li2S	mp-557142	P6_3/mmc	0.1750	1.5052	endobondic
Li3N*	mp-2341	P6_3/mmc	0.0039	0.6165	endobondic
LiCl*	mp-22905	Fm-3m	0.0243	2.8030	endobondic
LiCl	mp-1185319	P6_3mc	0.0039	2.5005	endobondic
LiF*	mp-1138	Fm-3m	0.0000	4.4673	endobondic
LiF	mp-1009009	Pm-3m	0.2875	4.5131	endobondic
LiP5*	mp-2412	Pna2_1	0.0077	0.9043	endobondic
LiP5	mp-32760	Pna2_1	0.0963	0.8692	endobondic
MgCl2*	mp-23210	R-3m	0.0000	0.1568	endobondic
MgCl2	mp-569626	Ama2	0.1991	-0.0862	exobondic
MgF2*	mp-1072956	Pnnm	0.0025	0.0015	endobondic
MgF2	mp-1062842	Amm2	0.2636	0.0229	endobondic
MgO*	mp-549706	P6_3mc	0.1180	-1.4179	exobondic
MgO	mp-1009127	Pm-3m	0.7484	-1.0763	exobondic
MgS*	mp-1315	Fm-3m	0.0000	-0.3401	exobondic
MgS	mp-13032	F-43m	0.0343	-0.4163	exobondic
Mn2O7*	mp-28338	P2_1/c	0.3346	-1.4687	exobondic
MnO2*	mp-510408	P4_2/mnm	0.0464	-0.5274	exobondic
Na2Cl	mp-1077015	Cmmm	0.2391	-0.3965	exobondic
Na2O*	mp-2352	Fm-3m	0.0000	-1.3057	exobondic
Na2O	mp-1975297	C2/m	0.0981	-1.5619	exobondic
NaN3*	mp-22003	R-3m	0.0017	-1.5201	exobondic
NaN3	mp-1065265	C2/m	1.0162	-5.4248	exobondic
NaS*	mp-409	P-62m	0.0023	-0.4911	exobondic
NaS	mp-1180106	C2/m	1.0061	-2.3999	exobondic
PbN6*	mp-667338	Pnma	0.1093	-2.5257	exobondic
Rb2O*	mp-1394	Fm-3m	0.0032	0.7235	endobondic
Rb2O	mp-2023482	P-3m1	0.0479	-0.7592	exobondic
RbCl*	mp-23299	Pm-3m	0.0298	1.7924	endobondic
RbF*	mp-2064	Pm-3m	0.0516	3.9244	endobondic
RbF	mp-11718	Fm-3m	0.0000	1.9497	endobondic
RbN3*	mp-581833	P4/mmm	0.0193	-1.3478	exobondic
RbN3	mp-1219567	Amm2	0.7966	-3.5636	exobondic
RbP*	mp-1179688	C2/m	0.9879	-1.9315	exobondic
RbP	mp-1058499	Immm	0.9882	-1.9104	exobondic
RbS*	mp-558071	Immm	0.0019	-0.3217	exobondic
RbS	mp-1179684	C2/m	0.9975	-2.7347	exobondic
S2N*	COD:4031496	P4_2nm	–	-0.9767	exobondic
SN*	COD:7017102	P2_1/c	–	-1.3963	exobondic

Formula	mp-id / COD	Space group	E_{hull} (eV/at)	ΔICOHP/at. (eV)	Label
Sr3P4*	mp-14288	Fdd2	0.0000	-1.0472	exobondic
Sr3P4	mp-682953	P1	0.0081	-1.0793	exobondic
SrF2*	mp-1102240	Pnma	0.0646	-0.5421	exobondic
SrF2	mp-1102911	Pnma	0.0267	-0.1481	exobondic
SrN*	mp-1058586	Fm-3m	0.4343	-4.5395	exobondic
SrO2*	mp-726705	Pnma	0.0096	-1.9205	exobondic
SrO2	mp-2763179	Pnma	0.0110	-1.7522	exobondic
SrS*	mp-10627	Pm-3m	0.2996	-0.1088	exobondic
TiO2*	mp-1439	Pbcn	0.0045	-0.6461	exobondic
TiO2*	mp-9173	Pnma	0.0575	-0.6509	exobondic
TiO2*	mp-430	P2_1/c	0.0395	-0.6267	exobondic

G Extension Campaign: Endobondic Reactions

The subset of case-1 decomposition reactions ($A_aB_b \to a\,A + b\,B$, standard-state elemental references, balanced by `reaction_icohp.py`) labeled **endobondic** (ΔICOHP ≥ 0, products − reactants convention — note this is the sign *opposite* `reaction_icohp.py`'s own `delta_icohp_per_atom` column reported elsewhere in this project; see `delta.py`'s docstring).

Table 6: Decomposition-into-elements (case 1) reactions labeled **endobondic** (ΔICOHP $\geq$ 0, products − reactants) among the 163 compounds of the `extension` campaign. * = experimental starting compound.

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	ΔICOHP/at. (eV)
0.3333 Be3N2 → Be + 0.3333 N2	mp-1017550	0.2081	0.3960
BeCl2* → Be + Cl2	mp-23267	0.0087	1.8204
BeF2* → Be + F2	mp-15951	0.0004	3.1514
BeO* → Be + 0.5 O2	mp-7599	0.0153	2.7864
BeO → Be + 0.5 O2	mp-1778	0.0070	4.4391
0.5 Cs2S* → Cs + 0.5 S	mp-1077814	0.0667	0.9554
CsCl* → Cs + 0.5 Cl2	mp-573697	0.0071	0.8830
CsF* → Cs + 0.5 F2	mp-8455	0.1088	3.8464
CsO2* → Cs + O2	mp-1441	0.0349	0.0666
CsO2 → Cs + O2	mp-1096936	0.0940	0.7743
KCl* → K + 0.5 Cl2	mp-23289	0.0776	0.7616
0.5 Li2O* → Li + 0.25 O2	mp-1960	0.0000	2.3366
0.5 Li2O → Li + 0.25 O2	mp-755894	0.0844	1.9481
0.5 Li2S* → Li + 0.5 S	mp-1125	0.0615	1.8505
0.5 Li2S → Li + 0.5 S	mp-557142	0.1750	1.5052
0.3333 Li3N* → Li + 0.1667 N2	mp-2341	0.0039	0.6165
LiCl* → Li + 0.5 Cl2	mp-22905	0.0243	2.8030
LiCl → Li + 0.5 Cl2	mp-1185319	0.0039	2.5005
LiF* → Li + 0.5 F2	mp-1138	0.0000	4.4673
LiF → Li + 0.5 F2	mp-1009009	0.2875	4.5131
LiP5* → Li + 5 P	mp-2412	0.0077	0.9043
LiP5 → Li + 5 P	mp-32760	0.0963	0.8692
MgCl2* → Mg + Cl2	mp-23210	0.0000	0.1568
MgF2* → Mg + F2	mp-1072956	0.0025	0.0015
MgF2 → Mg + F2	mp-1062842	0.2636	0.0229
0.5 Rb2O* → Rb + 0.25 O2	mp-1394	0.0032	0.7235
RbCl* → Rb + 0.5 Cl2	mp-23299	0.0298	1.7924
RbF* → Rb + 0.5 F2	mp-2064	0.0516	3.9244
RbF → Rb + 0.5 F2	mp-11718	0.0000	1.9497

H Extension Campaign: Exobondic Reactions

The complementary **exobondic** subset (ΔICOHP $<$ 0), same convention and provenance as Appendix G.

Table 7: Decomposition-into-elements (case 1) reactions labeled **exobondic** (ΔICOHP $<$ 0, products − reactants) among the 163 compounds of the `extension` campaign. * = experimental starting compound.

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	ΔICOHP/at. (eV)
0.2 Ba5P4* → Ba + 0.8 P	mp-17566	0.0000	-0.9590
BaCl2* → Ba + Cl2	mp-23199	0.0068	-0.2244
BaF3* → Ba + 1.5 F2	mp-1080821	0.0000	-0.8680
BaF3 → Ba + 1.5 F2	mp-1183303	0.0612	-0.2493
BaN2* → Ba + N2	mp-1102371	2.0568	-8.6248
BaO* → Ba + 0.5 O2	mp-1008500	0.0189	-0.3221
BaO → Ba + 0.5 O2	mp-1950989	0.0434	-1.6129
BaS3* → Ba + 3 S	mp-556296	0.0436	-0.6233
0.3333 Be3N2* → Be + 0.3333 N2	mp-18337	0.0000	-0.0746
0.3333 Be3P2* → Be + 0.6667 P	mp-567841	0.0000	-0.2111
0.3333 Be3P2 → Be + 0.6667 P	mp-1084771	0.1567	-0.1795
BeCl2 → Be + Cl2	mp-1172909	3.1688	-5.7807
BeF2 → Be + F2	mp-975644	2.5181	-0.6816
0.3333 Ca3N2* → Ca + 0.3333 N2	mp-1047	0.0167	-4.4881
0.3333 Ca3N2* → Ca + 0.3333 N2	mp-844	0.0000	-4.4374
0.3333 Ca3N2 → Ca + 0.3333 N2	mp-1013524	0.5112	-4.4898
CaCl2* → Ca + Cl2	mp-23214	0.0011	-0.3199
CaCl2 → Ca + Cl2	mp-1120739	0.0422	-0.2208
CaF2* → Ca + F2	mp-10464	0.0346	-0.2227
CaF2 → Ca + F2	mp-554355	0.2403	-0.1814
CaN* → Ca + 0.5 N2	mp-1058549	0.3982	-4.9183
CaO* → Ca + 0.5 O2	mp-1079707	0.2098	-2.2127
CaO* → Ca + 0.5 O2	mp-2605	0.0000	-0.9525
CaO → Ca + 0.5 O2	mp-1181903	2.7281	-2.9971
CsN3* → Cs + 1.5 N2	mp-510557	0.0000	-2.2903
CsN3 → Cs + 1.5 N2	mp-1225892	0.0113	-1.6664
CsP7* → Cs + 7 P	mp-1201790	0.0000	-0.2978
CsP7 → Cs + 7 P	mp-1213206	0.0001	-0.2667
0.5 K2O* → K + 0.25 O2	mp-971	0.0000	-0.4357
0.5 K2O → K + 0.25 O2	mp-1223784	0.2000	-1.5638
0.5 K2S* → K + 0.5 S	mp-1022	0.0000	-0.3359
KF3* → K + 1.5 F2	mp-1544557	0.0125	-1.1511
KF3 → K + 1.5 F2	mp-2749823	0.0119	-1.1505
KN3 → K + 1.5 N2	mp-636056	1.0370	-5.9991
KP* → K + P	mp-1058315	1.0080	-2.1600
KP → K + P	mp-1057787	1.0092	-2.1366
MgCl2 → Mg + Cl2	mp-569626	0.1991	-0.0862
MgO* → Mg + 0.5 O2	mp-549706	0.1180	-1.4179
MgO → Mg + 0.5 O2	mp-1009127	0.7484	-1.0763
MgS* → Mg + S	mp-1315	0.0000	-0.3401
MgS → Mg + S	mp-13032	0.0343	-0.4163
0.5 Mn2O7* → Mn + 1.75 O2	mp-28338	0.3346	-1.4687
MnO2* → Mn + O2	mp-510408	0.0464	-0.5274
0.5 Na2Cl → Na + 0.25 Cl2	mp-1077015	0.2391	-0.3965
0.5 Na2O* → Na + 0.25 O2	mp-2352	0.0000	-1.3057
0.5 Na2O → Na + 0.25 O2	mp-1975297	0.0981	-1.5619

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	ΔICOHP/at. (eV)
NaN3* → Na + 1.5 N2	mp-22003	0.0017	-1.5201
NaN3 → Na + 1.5 N2	mp-1065265	1.0162	-5.4248
NaS* → Na + S	mp-409	0.0023	-0.4911
NaS → Na + S	mp-1180106	1.0061	-2.3999
PbN6* → Pb + 3 N2	mp-667338	0.1093	-2.5257
0.5 Rb2O → Rb + 0.25 O2	mp-2023482	0.0479	-0.7592
RbN3* → Rb + 1.5 N2	mp-581833	0.0193	-1.3478
RbN3 → Rb + 1.5 N2	mp-1219567	0.7966	-3.5636
RbP* → Rb + P	mp-1179688	0.9879	-1.9315
RbP → Rb + P	mp-1058499	0.9882	-1.9104
RbS* → Rb + S	mp-558071	0.0019	-0.3217
RbS → Rb + S	mp-1179684	0.9975	-2.7347
0.5 S2N* → S + 0.25 N2	COD:4031496	–	-0.9767
SN* → S + 0.5 N2	COD:7017102	–	-1.3963
0.3333 Sr3P4* → Sr + 1.333 P	mp-14288	0.0000	-1.0472
0.3333 Sr3P4 → Sr + 1.333 P	mp-682953	0.0081	-1.0793
SrF2* → Sr + F2	mp-1102240	0.0646	-0.5421
SrF2 → Sr + F2	mp-1102911	0.0267	-0.1481
SrN* → Sr + 0.5 N2	mp-1058586	0.4343	-4.5395
SrO2* → Sr + O2	mp-726705	0.0096	-1.9205
SrO2 → Sr + O2	mp-2763179	0.0110	-1.7522
SrS* → Sr + S	mp-10627	0.2996	-0.1088
TiO2* → Ti + O2	mp-1439	0.0045	-0.6461
TiO2* → Ti + O2	mp-9173	0.0575	-0.6509
TiO2* → Ti + O2	mp-430	0.0395	-0.6267